SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I AB. Sai Saranya (1921260020) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Saranya.ab

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	2	1
Accuracy	10%	1	1	2	1
Clarity	5%	1	1	1	1

Total Marks 37/40 ***Signature of the Faculty***

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-struct ured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization , incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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Course Code: CSA0709

37/40

CO1 AT1

1. File transfer : segments, frames, overhead, time

Given :

- File = 150 MB
- Bandwidth = 50 Mbps
- segment = 1500 byte
- Data link header + trailer = 40 bytes per frame

(1MB = 1,000,000) bytes

To find :

1. Total number of segments & frames transmitted
2. Total transmission overhead
3. Total transmission time

Ans

Step 1 - To calculate total no of segments

$$150\ MB = 150 \times 10^6 \text{ bytes}$$

Formula - Number of segments = $\frac{150 \times 10^6}{1500}$

Number of segments = 100,000 segments.

Step 2 - To calculate the total no of Frames

Data link layer encapsulates each segments into frames.

∴ Total number of Frames = 100,000. frames

Step 3: Calculate the total Data link layer overhead

Total overhead = Number of Frames

= 100,000 x 40

= 4,000,000 bytes

To convert it into MB : $\frac{4,000,000}{10^6}$ = 4MB

Step 4 : Calculate the Total Transmission time

$150 \times 10^6 + 4 \times 10^6 = 154 \times 10^6$ bytes { Total data transmission }

To convert bytes into bits,

154,000,000 x 8 = 1,23,20,00,000 = 1232×10^6 bits

50 Mbps = 50×10^6 bps

Formula => $T = \frac{\text{Total bits}}{\text{bandwidth}} = \frac{1232 \times 10^6}{50 \times 10^6}$

T = 24.64 seconds

- Transport layer - divides data into segments, numbers them, uses acknowledgements and it retransmits lost segments.
- Data Link layer - packs each segment into a frame with MAC addresses and a CRC trailer. CRC detects bit errors on each link.
- Together - Data link makes every hop error-free and Transport guarantees the complete file reaches the server in order.

2. Given:

- Sliding window size = 6 frames
- User data per frame = 1024 bytes
- Total frames transmitted = 48

One frame is lost in every window

Step 1 Calculate the number of transmission windows

Formula - No of windows = $\frac{\text{Total frame}}{\text{Window size}}$

$= \frac{48}{6}$

$\Rightarrow$ 8 windows

Step 2 : Calculate the total events of retransmissio

Retransmissions = Windows x lost frame per window

= 8 x 1

= 8 Retransmissions

Step 3 : To calculate the user data transmitted per window

Data per window = Window time x Data per frame

= 6 x 1024

⇒ 6144 bytes

Flow control regulates the rate at which a sender transmits data so that the receiver is not overwhelmed.

⇒ In the sliding window protocol, the sender can transmit multiple frames without waiting for an acknowledgement after every frame. The receiver indicates how much data it can accept and the sender adjusts its transmission accordingly.

= 11,980 bytes

Step 3 - Calculate the number of fragments

$$\text{Fragments} = \frac{11,980}{1480}$$

$$= 8.095$$

$$\approx 9 \Rightarrow 9 \text{ fragments}$$

$11,980 - (8 \times 1480) = 140$ bytes

Step 4 - Calculate the total header overhead

Total header overhead = Number of Fragments X Header size

= 9 X 20

= 180 bytes

Total size of the fragments,

11,980 + 180 = 12,160 bytes

The Network layer uses IP fragmentation when a packet is too large for the MTV of a network link.

- Identification : Helps the destination identify fragments belonging to the same original packet.

The sliding window requires 8 windows and retransmissions under the stated assumption flow control improves communication efficiency by regulating data transmission according to the receiver's capacity.

3. Given:

- Original IP packet size = 12,000 bytes
- Maximum Transmission unit (MTV) = 1500 bytes
- IP header size per fragment = 20 bytes

To find:

1. Total number of fragments created
2. Total header overhead introduced

Step 1: To calculate the maximum data carried by each fragment.

Maximum fragment data = MTV − IP header

= 1500 − 20

= 1480 bytes

Step 2: Calculate the original payload size

Original payload = Packet size − Original header

= 12,000 − 20

8. Fragment offset - Indicates the position of each fragment in the original payload.

The Network layer provides the information required for reassembly while reliable re-transmission may be provided by a higher-layer protocol.

5. Session Layer Synchronisation

Given :

- Total video file size = 600 MB
- Checkpoint interval = 60 MB
- Failure occurs after 390 MB has been transfered

Step 1 - Calculate the no of complete checkpoints

$$\text{Checkpoint count} = \frac{\text{Data transfered}}{\text{Checkpoint interval}}$$

$$= \frac{390}{60} = 6.5$$

Only complete checkpoints are counted $\Rightarrow$ 6

The checkpoints occur at 60, 120, 180, 240, 300, 360 MB.

Step 2 Identify the last completed checkpoint

The last completed checkpoint before failure is

Retransmitted data = Transferred data - Last checkpoint

= 390 - 360

= 30 MB

The session layer manages communication & sessions between applications. Synchronisation checkpoints record progress during long data transfers.

The traditional OSI session layer describes synchronisation services, but actual checkpoint and recovery mechanisms.

Importance of checkpoints

→ They allow a transfer to resume from a saved point after failure.

→ They save time and network bandwidth

→ They improve the efficiency of large file transfers.